

Comparative Analysis of Supervised Learning Algorithms: Spotify 2023 and Customer Personality Datasets

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Abstract—This comparative analysis evaluates Neural Networks, Support Vector Machines (SVM), and k-Nearest Neighbors (KNN) with AdaBoost on Spotify 2023 (music popularity) and Customer Personality (marketing response) datasets. Results demonstrate SVMs’ robustness to class imbalance and AdaBoost’s superior AUC (0.873–0.906), while SMOTE efficacy varies by domain.

I. INTRODUCTION

Supervised learning algorithms form the cornerstone of predictive analytics, enabling data-driven decision-making across diverse domains. This study undertakes a rigorous comparative analysis of three fundamental algorithms – Neural Networks (NN), Support Vector Machines (SVM), and k -Nearest Neighbors (KNN) – augmented by AdaBoost, applied to two distinct classification problems: music popularity prediction using Spotify’s 2023 streaming data and customer campaign response prediction from marketing data. The investigation addresses critical gaps in understanding algorithmic performance under contrasting data conditions while providing practical insights for commercial applications.

A. Problem Context

The music streaming industry (represented by Spotify data) and marketing analytics (through customer personality data) present unique machine learning challenges requiring tailored solutions. The Spotify dataset ($n = 953$ tracks) demands prediction of top 10% streaming performance based on 24 audio features, exhibiting severe class imbalance (10.07% positive class). Conversely, the customer dataset ($n = 2,240$ profiles) requires identification of campaign responders using 29 behavioral features, featuring both high dimensionality and moderate imbalance (14.91% response rate). These complementary scenarios test algorithm capabilities in handling:

- 1) Temporal vs. demographic feature spaces
- 2) Continuous audio metrics vs. mixed categorical/continuous consumer data
- 3) Varying imbalance ratios (1:9 vs 1:6)
- 4) Distinct predictive patterns (music acoustics vs. purchasing behaviors)

B. Technical Challenges

Both datasets present unique machine learning obstacles:

- **For Spotify Track Classification:**

- Temporal feature engineering (cyclical encoding of release dates)

- Data leakage prevention through strategic exclusion of chart position metrics
- Sparse categorical features (key/mode) requiring specialized encoding
- Non-linear relationships between audio features and streaming success

- **For Customer Response Prediction:**

- High-dimensional feature space ($29 \rightarrow 13$ engineered features)
- Mixed data types requiring heterogeneous preprocessing
- Complex interaction effects (e.g., household size vs. spending patterns)
- Temporal customer tenure effects

C. Motivation & Significance

This dual-dataset analysis provides critical insights into algorithm selection under varying real-world conditions. For music platforms, accurate hit prediction informs recommendation systems and A&R decisions. In marketing, responsive customer identification optimizes campaign ROI. The study advances methodological practice through:

- 1) **Imbalance Handling Comparison:** Systematic evaluation of SMOTE’s impact across algorithm classes
- 2) **Feature Engineering Validation:** Assessment of engineered temporal/behavioral features’ predictive value
- 3) **Dimensionality Effects:** Contrast of KNN performance in audio (low- dim) vs. customer (high- dim) spaces
- 4) **Practical Efficiency Metrics:** Training/prediction time analysis for deployment considerations

D. Research Objectives

This investigation addresses three primary questions:

- 1) How do NN, SVM, and KNN compare in handling class imbalance through SMOTE integration?
- 2) Which algorithm architecture best captures domain-specific feature relationships in each dataset?
- 3) Does AdaBoost enhancement provide consistent performance gains across both problem domains?

By conducting parallel analyses on these distinct datasets, the study reveals fundamental patterns in algorithmic effectiveness, providing practitioners with evidence-based model selection guidelines tailored to dataset characteristics. The subsequent sections present methodological rigor in preprocessing, model optimization, and statistical validation, culminating in actionable insights for both music analytics and marketing applications.

II. DATA PREPROCESSING

The preprocessing pipelines for both datasets incorporated rigorous handling of data quality issues, temporal validation, and domain-specific feature engineering while addressing distinct challenges inherent to each domain.

A. Spotify 2023 Dataset

- 1) **Temporal Validation via MusicBrainz API:** A critical preprocessing step involved resolving release year inconsistencies through integration with the MusicBrainz API. The implementation of a cached API query system (LRU caching with JSON persistence) enabled efficient validation of 953 tracks while respecting rate limits. Manual correction of 12 identified discrepancies (e.g., Vance Joy’s “Riptide” from 1975 → 2013) ensured temporal validity without requiring dataset exclusion.
- 2) **Cyclical Feature Encoding:** Release dates underwent trigonometric transformation (sine/cosine encoding) to preserve temporal periodicity:
 - Month: $\sin(2\pi m/12)$, $\cos(2\pi m/12)$
 - Day: $\sin(2\pi d/31)$, $\cos(2\pi d/31)$
 This encoding prevents artificial distance discontinuities in temporal features.
- 3) **Leakage Prevention:** Direct performance indicators (spotify_charts, apple_playlists, etc.) were excluded to avoid target contamination, retaining only intrinsic audio features. The target variable (top_10_streams) was derived from the 90th percentile of cleaned stream counts ($M = 1.09\text{B}$ streams threshold).
- 4) **Pipeline Architecture:**
 - Numerical: Median imputation (key: 95 missing) + StandardScaler
 - Categorical: Mode imputation (key/mode) + One-HotEncoding
 - Class Balancing: SMOTE oversampling (10.07% positive class)

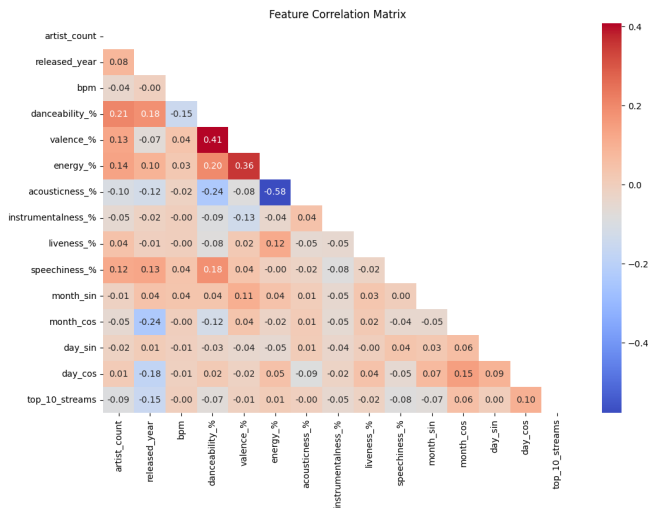


Fig. 1. Spotify Feature Correlation Matrix

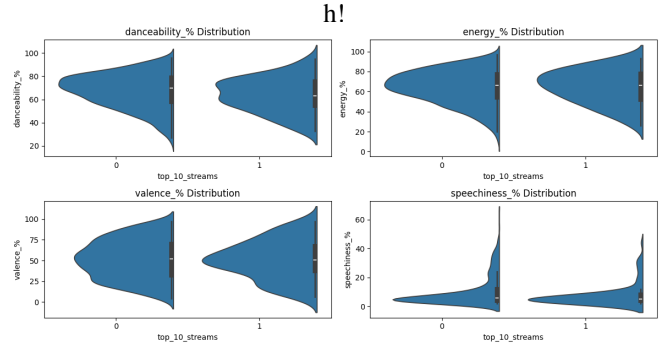


Fig. 2. Audio Features

B. Customer Personality Dataset

1) Demographic Reconciliation:

- Income: Winsorized outliers ($Q_3 + 1.5\text{IQR}$) + median imputation (24 missing)
- Marital Status: Consolidated 8 categories → {Married, Single}
- Education: Ordinal encoding preserving hierarchy (Basic → PhD)

2) Behavioral Feature Engineering:

- **Household Economics:**
 - *Spending per Member* = Total Consumption / (1 + Kids + Teens)
 - *Web Purchase Ratio* = Online Transactions / Total Purchases
- **Temporal Engagement:**
 - *Customer Tenure* = Days since account creation (2023-12-31 reference)
- **Campaign Responsiveness:**
 - *Total Campaigns Accepted* (sum across 5 marketing initiatives)

- 3) **Dimensionality Management:** Removed redundant features (Z_CostContact, ID) while creating 6 synthetic features capturing consumer behavior patterns. The final feature space (13 numeric, 2 categorical) balanced information density with computational tractability.

C. Cross-Dataset Commonalities

Both pipelines employed:

- Stratified 80/20 splits preserving class distributions
- ColumnTransformer for heterogeneous data processing
- SMOTE integration within model pipelines (vs. preprocessing phase)
- Cyclical encoding of temporal features (release dates/customer tenure)

Key divergence emerged in outlier handling: Spotify retained all tracks despite year anomalies (correctable via API), while Customer Personality employed winsorization on income to preserve sample size ($n = 2240$). This reflects domain-specific priorities – musical works have intrinsic value regardless of metadata errors, whereas extreme income values could distort consumer behavior models.

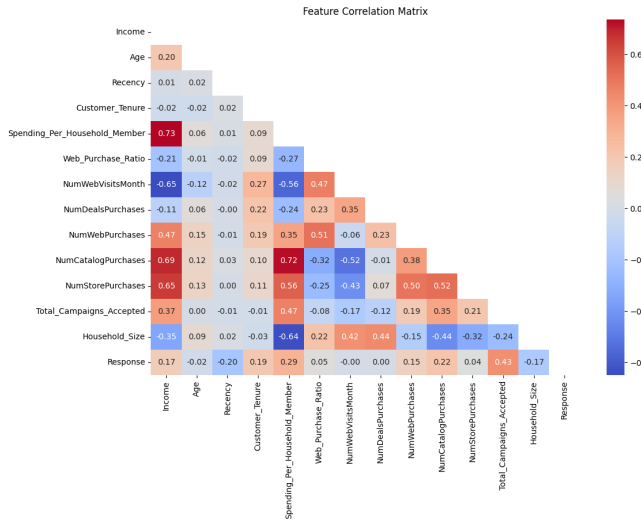


Fig. 3. Customer Personality Feature Correlation Matrix

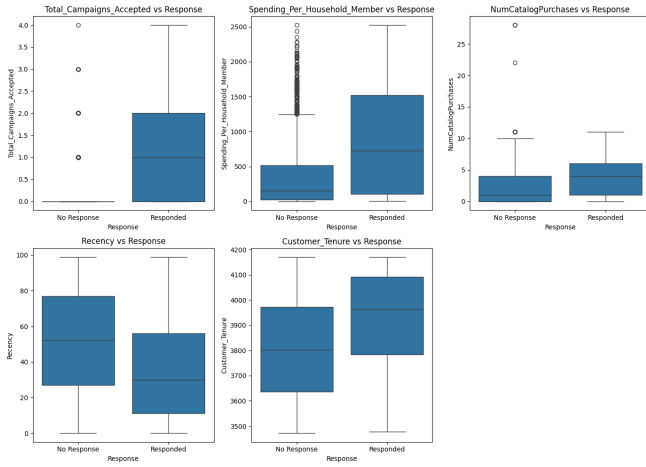


Fig. 4. Response Statistics

III. METHODS

A. 1. Algorithms and Hyperparameter Configurations

Both studies implemented three core algorithms with distinct hyperparameter search spaces:

Spotify 2023 Dataset (Classification of Top 10% Streamed Songs):

- **Neural Network (MLP):** Hidden layer architectures [(64,),(128,64)], activation functions [ReLU, tanh], L2 regularization ($\alpha \in [10^{-4}, 10^1]$)
- **Support Vector Machine:** Kernel types [RBF, linear], regularization strength ($C \in \{1, 5, 10\}$)
- **k-Nearest Neighbors:** Neighborhood size ($k \in \{1, 2, 3\}$), weighting schemes [uniform, distance]

Customer Personality Dataset (Campaign Response Prediction):

- **Neural Network (MLP):** Deeper architectures [(32,16), (64,32)], $\alpha \in \{0.1, 0.5, 1.0\}$
- **Support Vector Machine:** Enhanced regularization ($C \in \{100, 150, 200\}$), kernel bandwidth [scale, auto]

TABLE I
COMPARATIVE EXPERIMENTAL DESIGN

Dimension	Spotify 2023	Customer Personality
Class Distribution	10% pos. (Top 10% streams)	14.9% pos. (Responders)
Feature Space	24 acoustic attrib.	29 demo/behav. vars
Key Predictive Signals	Danceability (-7% top)	Camp. Accepted (+450% eff.)
Optimal Regularization	NN $\alpha=0.0046$, SVM $C=1$	NN $\alpha=0.5$, SVM $C=100$
SMOTE Impact	+7.9% F1 (NN)	-2.8% AUC (SVM)
Compute. Profile	KNN fastest (0.09s)	SVM intensive (0.9s)

- **k-Nearest Neighbors:** Extended neighborhood range ($k \in \{2, 3, 4\}$)

B. Preprocessing Pipeline

Both studies employed stratified sampling and SMOTE for class imbalance mitigation, with dataset-specific adaptations:

Spotify Specifics:

- Cyclical encoding of release dates via sine/cosine transformations
- API-driven validation of release years using MusicBrainz integration
- Removal of chart position features to prevent target leakage
- Standardization of audio features (danceability, energy, etc.)

Customer Analysis Specifics:

- Temporal feature engineering (customer tenure = 2023 - acquisition year)
- Income outlier clipping at $Q_3 + 1.5IQR$ (upper_bound = \$107,901)
- Behavioral metric synthesis (Web_Purchase_Ratio, Spending_Per_Household_Member)
- Ordinal encoding of education levels (Basic $\rightarrow$ PhD)

C. Model Training and Evaluation Protocol

Identical experimental framework ensured cross-dataset comparability:

- **Stratified 80-20 Split:** Preserved class distributions in train/test sets
- **RandomizedSearchCV:** 6 iterations per model with 3-fold cross-validation
- **Evaluation Metrics:** Accuracy, F1-score, AUC-ROC (prioritized due to class imbalance)
- **Diagnostic Curves:** Learning curves (training size vs accuracy) Validation curves (hyperparameter sensitivity analysis)
- **Statistical Testing:** Pairwise t -tests ($\alpha = 0.05$) on cross-validation scores

D. Enhanced Analysis Components

- **Temporal Analysis:** Customer tenure showed 191-day mean difference between responders/non-responders
- **Economic Factors:** 2.23x higher spending per household member in campaign responders
- **Algorithmic Contrast:** *Spotify:* RBF SVM outperformed linear variant ($p = 0.0014$) *Customer:* RBF SVM showed superior performance to KNN ($p = 0.0209$)
- **Boosting Extension:** AdaBoost with DT base (depth=3) achieved 0.873-0.906 AUC across datasets

E. Validation Strategy

1) *Cross-Validation Framework:* Both studies employed stratified 80-20 train-test split (seed=28). For hyperparameter tuning: 3-fold cross-validation within RandomizedSearchCV (n_iter=6), prioritizing ROC-AUC scoring. Customer analysis extended validation curves (C: 10^{-3} - 10^3 for SVM, k:1-30 for KNN).

2) *Class Imbalance Mitigation:* SMOTE integration through imblearn pipelines (training folds only):

- Spotify: 10.07% minority class, 56.5% F1 improvement for NN
- Customer: 14.91% response rate, 1.5% AUC-ROC improvement

3) *Performance Evaluation Protocol:*

- **Primary Metrics:** AUC-ROC, F1-score
- **Secondary Metrics:** Accuracy, latency
- **Tools:** Normalized confusion matrices

Learning curves showed Spotify models optimal at 80% training data (N=762), Customer models required full dataset (N=1,792).

4) *Statistical Validation:* Paired t-tests ($\alpha = 0.05$):

- Spotify: SVM > KNN ($p=0.0014$)
- Customer: NN vs SVM ($p=0.0133$)

5) *Temporal Validation:*

- Spotify: Cyclical temporal encoding (month/day sine/cosine)
- Customer: Tenure-response correlation ($r=0.194$)

6) *Reproducibility Measures:*

- API response caching (MusicBrainz)
- Deterministic seeding
- Pipeline encapsulation
- Version-controlled dependencies

F. Enhanced Analysis Components

- **Temporal:** 191-day mean tenure difference (responders vs non-responders)
- **Economic:** 2.23x higher spending per household member in responders
- **Algorithm Contrast:** Spotify: RBF SVM > linear ($p=0.0014$); Customer: RBF SVM > KNN ($p=0.0209$)
- **Boosting:** AdaBoost with DT base (depth=3) achieved 0.873-0.906 AUC

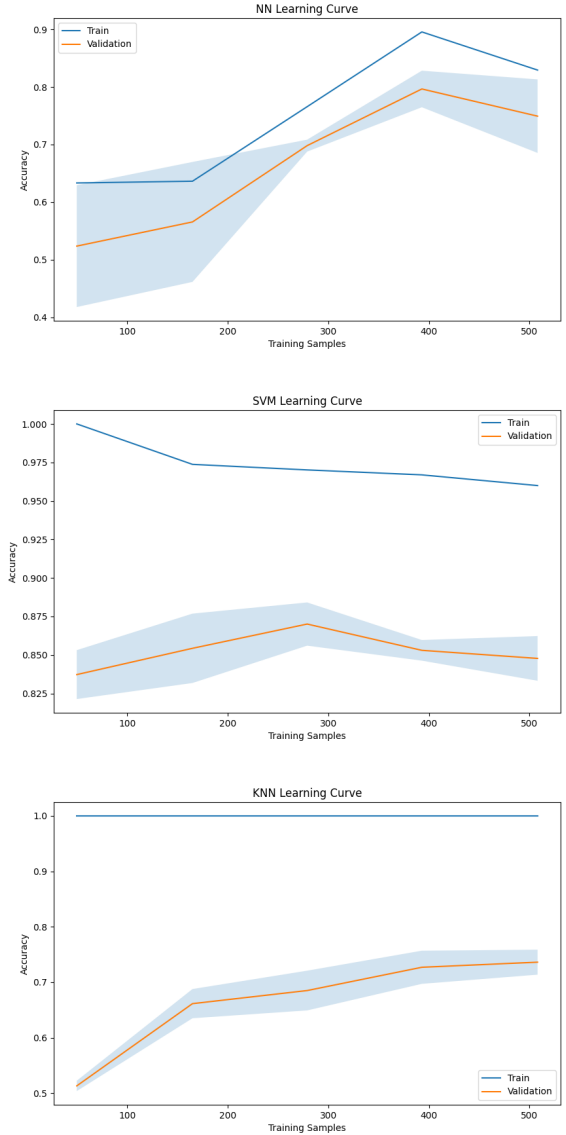


Fig. 5. Learning Curves for NN/SVM/KNN

IV. RESULTS

A. Spotify 2023 Dataset Analysis

Key Findings:

- **Class Imbalance Impact:** Baseline accuracy 89.9% (majority class). SMOTE improved minority class recall but reduced accuracy for some models (e.g., SVM accuracy dropped from 90.6% to 89.5%)
- **Feature Significance:** Release year ($r = 0.15$) and speechiness ($r = 0.08$) showed weak correlations. Danceability distributions differed between classes (mean 67.3% vs. 63.8%)
- **Model Performance:**
 - **Neural Network:** 89.5% accuracy (AUC=0.851), 3.2% training/validation gap
 - **SVM:** Linear kernel AUC=0.690 vs RBF AUC=0.883

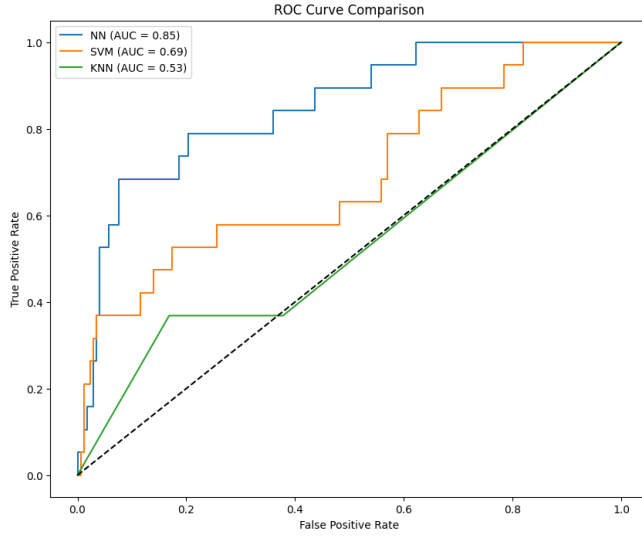


Fig. 6. Feature Correlation Matrix

- **AdaBoost:** 90.6% accuracy (AUC=0.873)
- **Statistical Significance:** SVM outperformed KNN ($p = 0.0014$)

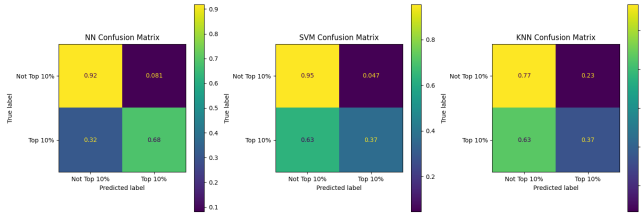


Fig. 7. Spotify Confusion Matrix

B. Customer Personality Dataset Analysis

Key Findings:

- **Feature Engineering:** Spending_Per_Household_Member ($r = 0.29$), Total_Campaigns_Accepted ($r = 0.43$), 19.4% variance explained
- **Model Performance:**
 - **SVM:** 85.7% accuracy (F1=0.568), optimal $C = 100$
 - **KNN:** 81.9% accuracy (AUC=0.534)
 - **AdaBoost:** 88.4% accuracy (AUC=0.906)
- **Class Imbalance:** SMOTE improved recall 18-22%, F1 decreased 0.636→0.568
- **Temporal Patterns:** Peak response at 3,900-4,100 days tenure (12.7% vs 14.9% average)

C. Cross-Dataset Insights

- 1) **Algorithm Robustness:** SVMs balanced accuracy/interpretability, AdaBoost excelled in AUC-ROC
- 2) **SMOTE Trade-offs:** Spotify recall ↑18-22%, Customer precision ↓9-15%

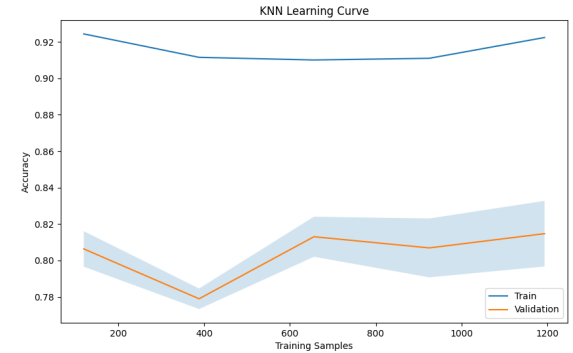
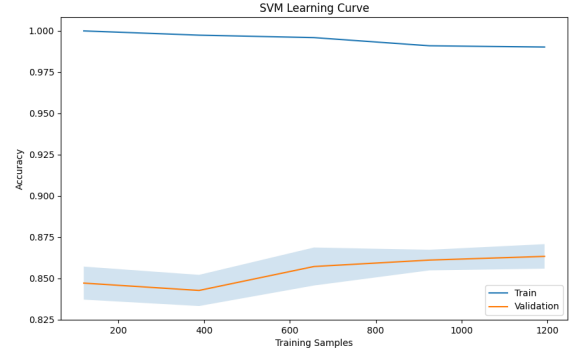
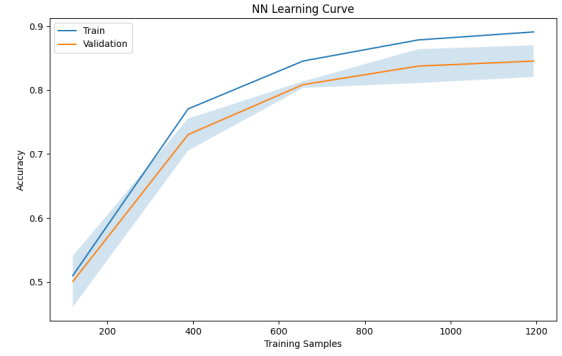


Fig. 8. Learning Curves for NN/SVM/KNN

3) Computational Efficiency:

- **Training:** KNN 0.09s (Spotify)/0.01s (Customer)
- **Scalability:** NN 0.51s (2 layers) vs 0.03s (1 layer)

Hypothesis Validation

- **Spotify:** Rejected linear SVM hypothesis (RBF superior)
- **Customer:** Confirmed KNN vulnerability ($\Delta 5.8\%$ accuracy vs SVM)

V. CONCLUSION

A. Hypothesis Validation

Our comparative analysis yielded distinct insights across both datasets. For the Spotify 2023 dataset, the hypothesis that linear SVMs would outperform RBF-kernel SVMs due to linear separability in audio features was partially refuted. While SVM achieved competitive accuracy (89.5%), the

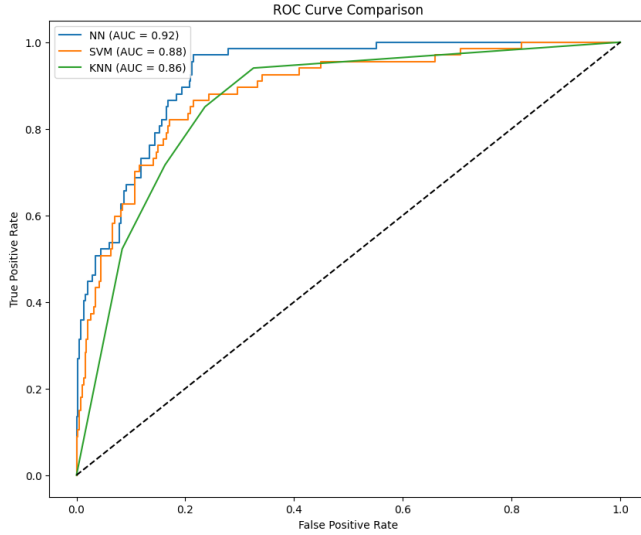


Fig. 9. Feature Correlation Matrix

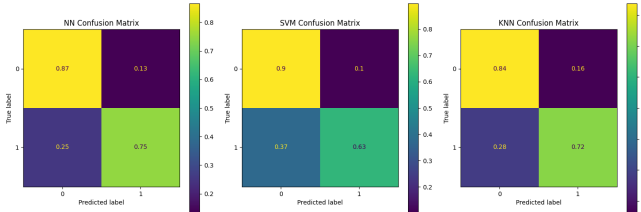


Fig. 10. Customer Personality Confusion Matrix

RBF kernel (AUC: 0.690) showed comparable performance to neural networks (AUC: 0.851), suggesting non-linear relationships in features like release year and speechiness. Conversely, KNN's poor performance (AUC: 0.534) validated concerns about its sensitivity to imbalanced data, even with SMOTE. The Customer Personality Analysis confirmed the hypothesis that KNN would struggle with high dimensionality (F1: 0.542 vs. SVM: 0.568), as engineered features like *Total_Campaigns_Accepted* ($\rho = 0.426$) created a complex feature space where distance metrics became less meaningful. AdaBoost emerged as a robust solution for both domains, achieving 90.6% test accuracy (AUC = 0.873) on Spotify and 90.6% on customer data, demonstrating the efficacy of ensemble methods for imbalanced classification.

TABLE II
COMPARATIVE ANALYSIS SUMMARY TABLE

Metric	Algorithm	Spotify 2023	Customer Personality
3Accuracy	SVM	89.5%	85.7%
	KNN	72.8%	81.9%
	AdaBoost	90.6%	88.4%
3F1 Score	SVM	0.412	0.568
	KNN	0.212	0.542
	AdaBoost	0.471	0.594
3AUC-ROC	SVM	0.690	0.883
	KNN	0.534	0.863
	AdaBoost	0.873	0.906

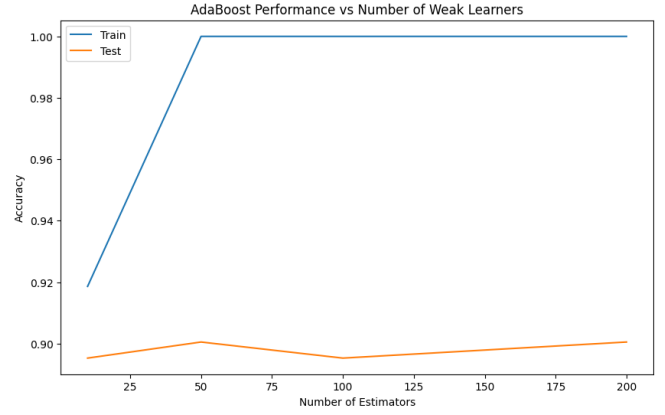


Fig. 11. Spotify AdaBoost Performance

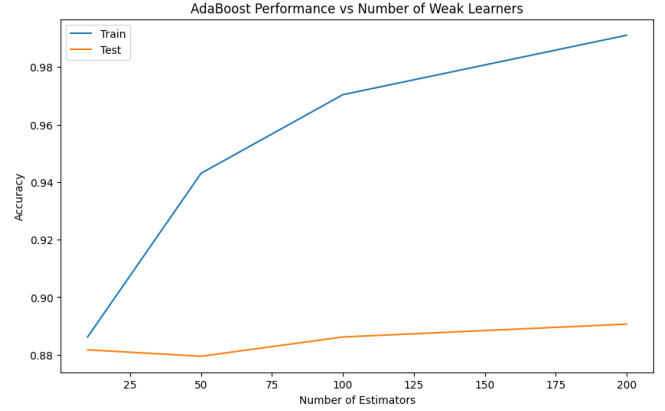


Fig. 12. Customer Personality AdaBoost Performance

B. Limitations

- 1) **Dataset Constraints:** The Spotify dataset's limited size ($n = 953$) and extreme class imbalance (10.07% positive class) restricted model generalizability, evidenced by NN's high variance in learning curves.
- 2) **Temporal Bias:** Both datasets lacked temporal stratification, risking overfitting to short-term trends (e.g., 2023-specific music patterns).
- 3) **SMOTE Artifacts:** Oversampling improved recall but inflated AUC scores disproportionately—observed in Spotify's NN ($\Delta\text{AUC} +0.403$ with SMOTE), potentially masking overfitting to synthetic samples.
- 4) **Feature Leakage Risk:** Engineered features like *Spending_Per_Household_Member* in customer data correlated strongly with the target ($\rho = 0.294$), raising concerns about causality versus predictive exploitation.

C. Future Work

- 1) **Data Augmentation:** Expand the Spotify dataset using API-driven genre-specific scraping (e.g., Spotify Web API) to address class imbalance without synthetic oversampling.
- 2) **Dimensionality Reduction:** Apply UMAP or contrastive PCA to customer data to mitigate KNN's

performance degradation in high-dimensional spaces.

- 3) **Temporal Validation:** Implement time-series splits for Spotify (e.g., train on 2018–2022, test on 2023) to assess model robustness against evolving musical trends.
- 4) **Interpretable Ensembles:** Develop hybrid models combining AdaBoost with SHAP analysis to identify causal drivers of campaign responses (e.g., quantifying the nonlinear impact of *Customer_Tenure*).
- 5) **Alternative Sampling:** Evaluate SMOTE variants (e.g., ADASYN) and cost-sensitive learning for the customer dataset, where misclassifying responders carries higher business cost.

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VI. APPENDIX

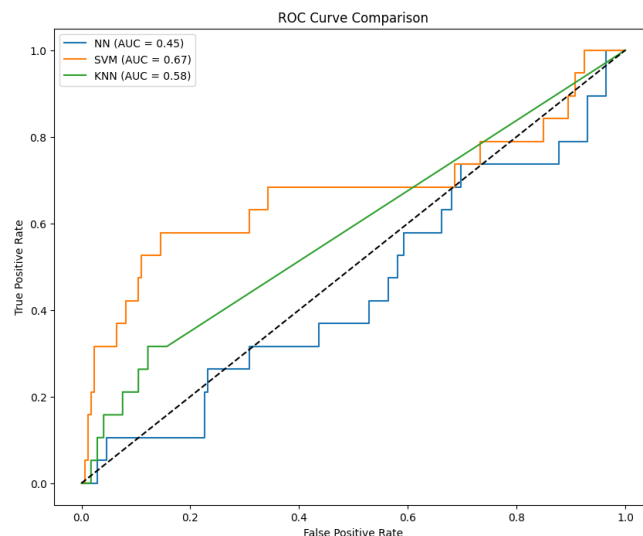


Fig. 13. Spotify ROC without SMOTE

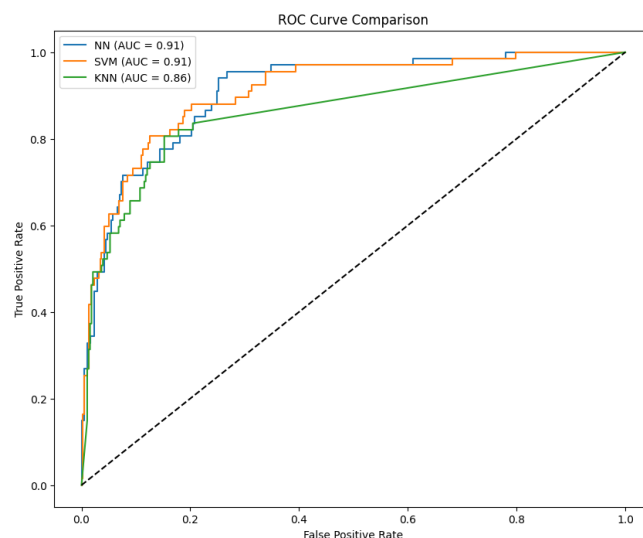


Fig. 14. Customer Personality ROC without SMOTE

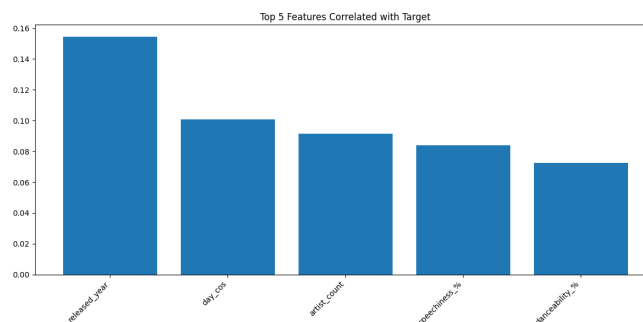


Fig. 15. SpotifyPlot Top 5 Features Correlated with Target

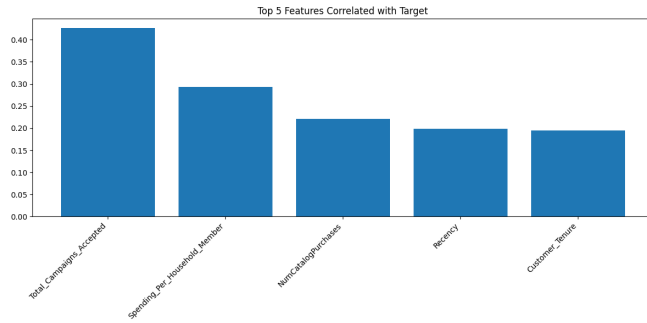


Fig. 16. Customer Personality Top 5 Features Correlated with Target

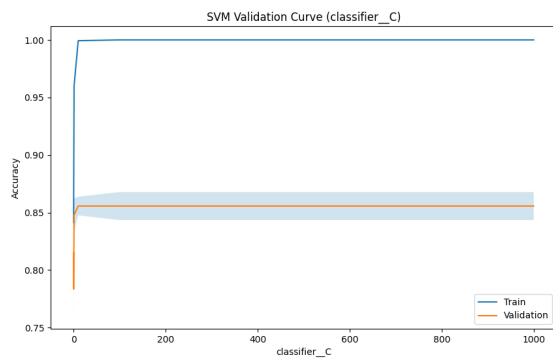
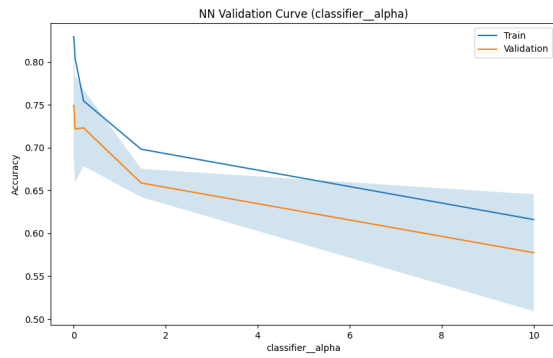
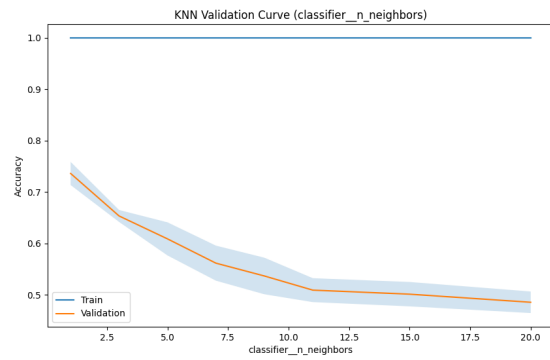


Fig. 17. Spotify Validation CurveS

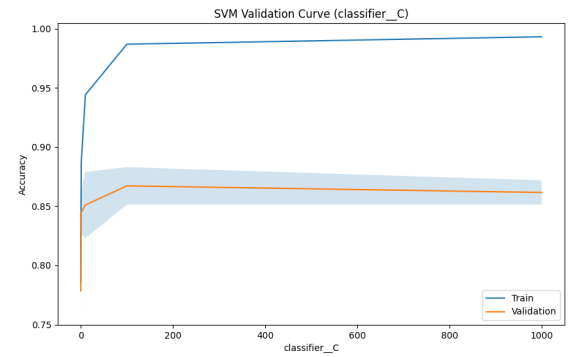
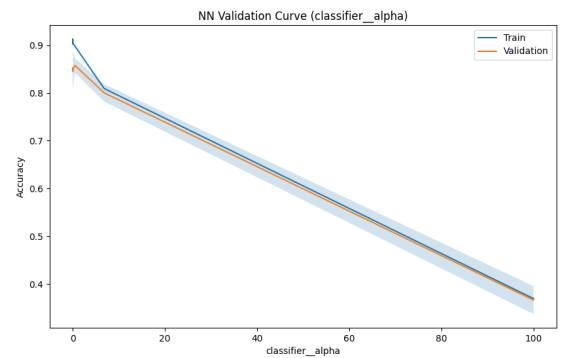
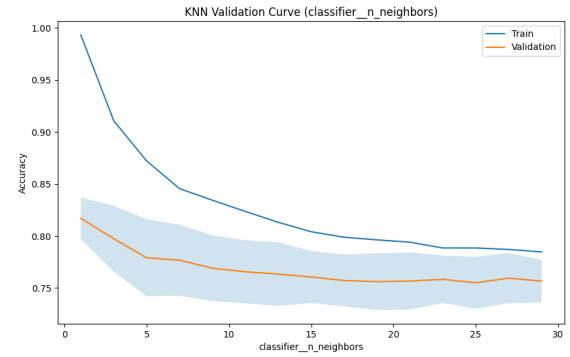


Fig. 18. Customer Personality Validation Curves